Personal Finance Tracker

Test Case Report & Bug Report

Project: Personal Finance Tracker
Document: Test Case Report & Bug Report
Tester: Bassam Ashraf
Date: March 2026
Version: 1.0
Total TCs: 27 Test Cases (TC-001 – TC-027)
Total Bugs: 4 Bugs Found

Table of Contents

Table of Contents 2
1. Introduction 3
2. Test Environment 3
3. Test Execution Summary 3
4. Test Cases 4
4.1. Description Field 4
4.2. Amount Field 4
4.3. Type Field 4
4.4. Category Field 5
4.5. Date Field 5
4.6. Note Field 6
4.7. Transaction List 6
5. Bug Report 7
5.1. BUG-001 — Amount Above Maximum ($1,000,000.00) Is Accepted 8
5.2. BUG-002 — Date Below Minimum Boundary (1899-12-31) Is Accepted 9
5.3. BUG-003 — Date Above Maximum Boundary (2100-01-01) Is Accepted 10
5.4. BUG-004 — Notes Field Accepts More Than 200 Characters 11
5.5. BUG-005 — Long Text in Description or Notes Causes UI Layout Overflow 12
6. Recommendations 13

1. Introduction

This document presents the complete automated test case report and bug report for the Personal Finance Tracker v1.0 application. The application is a client-side web application built with HTML5, CSS3, and JavaScript (ES6+) that allows users to manage personal income and expense transactions with real-time balance calculations and localStorage persistence.

Testing was performed using the Selenium WebDriver automation framework with the Mocha test runner and Chai assertion library. All tests were executed in Google Chrome. A total of 27 test cases were executed across 7 test suites covering all input fields and core functional behaviour.

2. Test Environment

Application: Personal Finance Tracker v1.0
Test File: Test.js (TC-001 – TC-027)
Framework: Selenium WebDriver 4.x
Test Runner: Mocha 11.x
Assertions: Chai 6.x
Browser: Google Chrome (Latest)
Driver: ChromeDriver (auto-resolved via chromedriver npm)
OS: Windows
Spec Reference: Specification_Document.md

3. Test Execution Summary

Total Test Cases	Passed	Failed	Pass Rate
27	**23**	**4**	**85%**

Test Suite	Total TCs	Passed	Failed	Bugs Found
Description Field	5	5	0	1
Amount Field	6	5	1	1
Type Field	3	3	0	—
Category Field	2	2	0	—
Date Field	6	4	2	2
Note Field	3	2	1	2
Transaction List	2	2	0	—

4. Test Cases

4.1. Description Field

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-001	Description Field	Empty Description Field	Equivalence Partitioning	(empty)	Form rejected; browser flags field invalid; balance unchanged	**PASSED**
TC-002	Description Field	Whitespace-Only Description	Equivalence Partitioning	" " (single space)	Error "Description is required"; balance unchanged	**PASSED**
TC-003	Description Field	Minimum Length — 1 Character	Boundary Value Analysis	"a" (1 char)	Transaction accepted; description "a" in list	**PASSED**
TC-004	Description Field	Maximum Length — 100 Characters	Boundary Value Analysis	100-char string	Transaction accepted; full 100-char description displayed	**PASSED**
TC-005	Description Field	Exceeds Maximum — 101 Characters	Boundary Value Analysis	101-char string	Error "Description must be less than 100 characters"; no transaction added	**PASSED**

4.2. Amount Field

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-006	Amount Field	Negative Amount	Boundary Value Analysis	-50	Error "Please enter a valid positive amount"; no transaction added	**PASSED**
TC-007	Amount Field	Minimum Valid Amount — $0.01	Boundary Value Analysis	0.01	Transaction accepted; amount $0.01 displayed	**PASSED**
TC-008	Amount Field	Maximum Valid Amount — $999,999.99	Boundary Value Analysis	999999.99	Transaction accepted; amount $999,999.99 displayed	**PASSED**
TC-009	Amount Field	Exceeds Maximum — $1,000,000.00	Boundary Value Analysis	1000000.00	Error "Amount seems too large"; no transaction added	**FAILED**
TC-010	Amount Field	Three Decimal Places — 5.001	Equivalence Partitioning	5.001	Browser step-mismatch validation; no transaction added	**PASSED**
TC-011	Amount Field	Empty Amount Field	Equivalence Partitioning	(empty)	Browser flags required field invalid; no transaction added	**PASSED**

4.3. Type Field

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-012	Type Field	Empty Type Field	Equivalence Partitioning	(no selection)	Browser flags required field invalid; no transaction added	**PASSED**
TC-013	Type Field	Income Transaction Type	Equivalence Partitioning	"income"	Transaction added; balance and total income each increase by $100	**PASSED**
TC-014	Type Field	Expense Transaction Type	Equivalence Partitioning	"expense"	Transaction added; balance decreases by $100; total expenses increases by $100	**PASSED**

4.4. Category Field

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-015	Category Field	Category Field Empty (Optional)	Equivalence Partitioning	(no selection)	Transaction accepted; "No category" displayed	**PASSED**
TC-016	Category Field	Category with Special Character (&)	Equivalence Partitioning	"Bills & Utilities"	Transaction accepted; "Bills & Utilities" stored and displayed correctly	**PASSED**

4.5. Date Field

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-017	Date Field	Past Date — 2024-01-15	Equivalence Partitioning	2024-01-15	Transaction accepted; date stored as 2024-01-15	**PASSED**
TC-018	Date Field	Default Date (Today)	Equivalence Partitioning	(default — today)	Transaction accepted; date stored as today's date	**PASSED**
TC-019	Date Field	Minimum Valid Date — 1900-01-01	Boundary Value Analysis	1900-01-01	Transaction accepted; date stored as 1900-01-01	**PASSED**
TC-020	Date Field	Maximum Valid Date — 2099-12-31	Boundary Value Analysis	2099-12-31	Transaction accepted; date stored as 2099-12-31	**PASSED**
TC-021	Date Field	Below Minimum Date — 1899-12-31	Boundary Value Analysis	1899-12-31	Transaction rejected; date 1899-12-31 not stored	**FAILED**

TC-022	Date Field	Above Maximum Date — 2100-01-01	Boundary Value Analysis	2100-01-01	Transaction rejected; date 2100-01-01 not stored	**FAILED**

4.6. Note Field

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-023	Note Field	Minimum Note Length — 1 Character	Boundary Value Analysis	"b" (1 char)	Transaction accepted; note "b" stored and displayed	**PASSED**
TC-024	Note Field	Maximum Note Length — 200 Characters	Boundary Value Analysis	200-char string	Transaction accepted; full 200-char note stored without truncation	**PASSED**
TC-025	Note Field	Exceeds Maximum Note — 201 Characters	Boundary Value Analysis	201-char string	Transaction rejected; 201-char note not stored	**FAILED**

4.7. Transaction List

TC ID	Suite	Test Case	Technique	Input	Expected Result	Status
TC-026	Transaction List	Reverse Chronological Order (3 Txns)	Functional Testing	3 sequential transactions	THIRD at position 0, SECOND at position 1, FIRST at position 2	**PASSED**
TC-027	Transaction List	Delete Last Transaction	Functional Testing	"delete me" transaction	Transaction removed; count decreases by 1	**PASSED**

5. Bug Report

5 bugs were identified during test execution. The table below summarises all bugs, followed by detailed reports with screenshots for each.

Bug ID	Title	Severity	TC	Component
BUG-001	Amount Above Maximum ($1,000,000.00) Is Accepted	**HIGH**	TC-009	Amount Field Validation
BUG-002	Date Below Minimum Boundary (1899-12-31) Is Accepted	**MEDIUM**	TC-021	Date Field Validation
BUG-003	Date Above Maximum Boundary (2100-01-01) Is Accepted	**MEDIUM**	TC-022	Date Field Validation
BUG-004	Notes Field Accepts More Than 200 Characters	**HIGH**	TC-025	Notes Field Validation
BUG-005	Long Text in Description or Notes Causes UI Layout Overflow	**MEDIUM**	TC-004, TC-025	Transaction List UI — Responsive Layout

5.1. BUG-001 — Amount Above Maximum ($1,000,000.00) Is Accepted

Bug ID	BUG-001
Severity	**HIGH**
Related TC	TC-009
Component	Amount Field Validation
Spec Reference	Section 3.2 — Amount Field: Maximum Value = $999,999.99

Steps to Reproduce:

1. Navigate to the application.
2. Enter "hello" in the Description field.
3. Enter 1000000.00 in the Amount field.
4. Select "Income" from the Type dropdown.
5. Select "Salary" from the Category dropdown.
6. Click "Add Transaction".

Expected Result	Actual Result
Form is rejected. Error message "Amount seems too large" is displayed. No transaction is added.	Application accepts $1,000,000.00 and adds the transaction to the list without any validation error.

Screenshot:

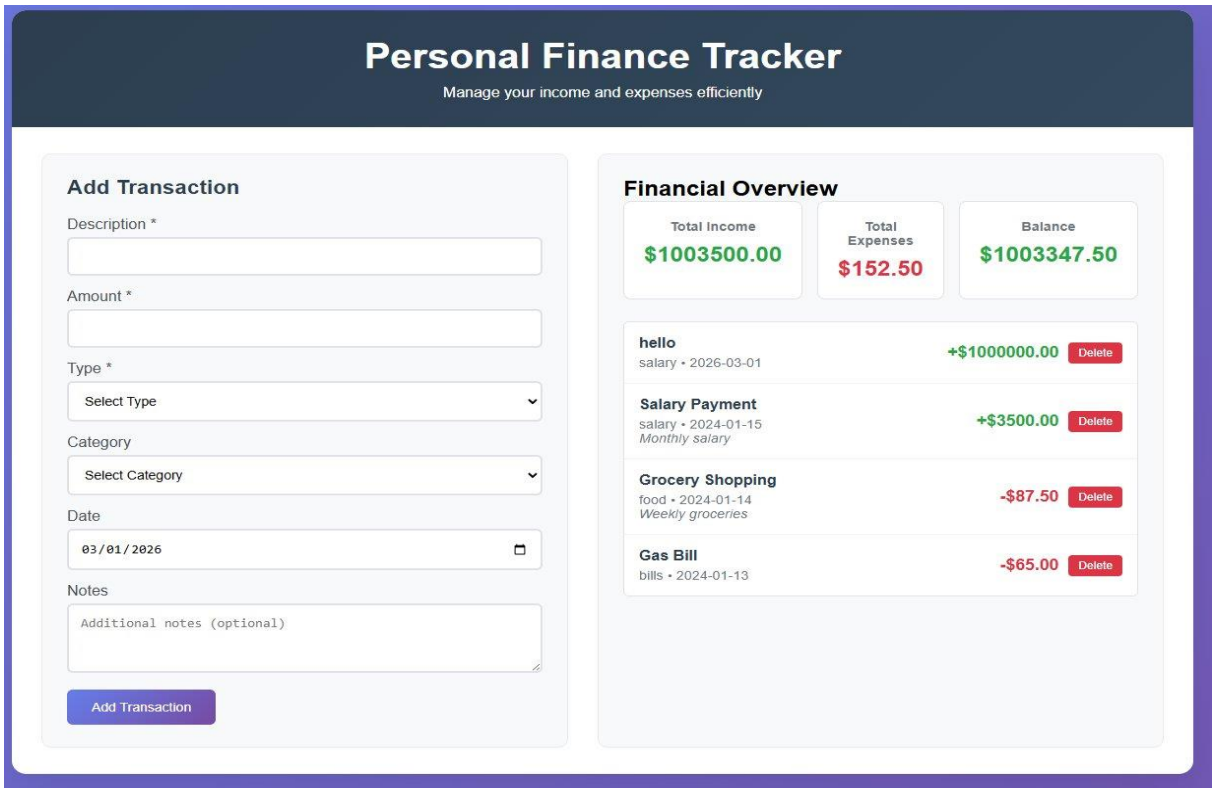

Figure 1 — BUG-001: Amount $1,000,000.00 accepted; transaction added showing +$1,000,000.00

5.2. BUG-002 — Date Below Minimum Boundary (1899-12-31) Is Accepted

Bug ID	BUG-002
Severity	**MEDIUM**
Related TC	TC-021
Component	Date Field Validation
Spec Reference	Section 3.5 — Date Field: Range = 1900-01-01 to 2099-12-31

Steps to Reproduce:

7. Navigate to the application.
8. Enter "a" in the Description field.
9. Enter 100 in the Amount field.
10. Select "Expense" from the Type dropdown.
11. Select "Shopping" from the Category dropdown.
12. Set the Date field to "1899-12-31" via JavaScript injection.
13. Click "Add Transaction".

Expected Result	Actual Result
Transaction is rejected. Date "1899-12-31" is out of the allowed range and should not be stored.	Application accepts the date 1899-12-31 and adds the transaction to the list.

Screenshot:

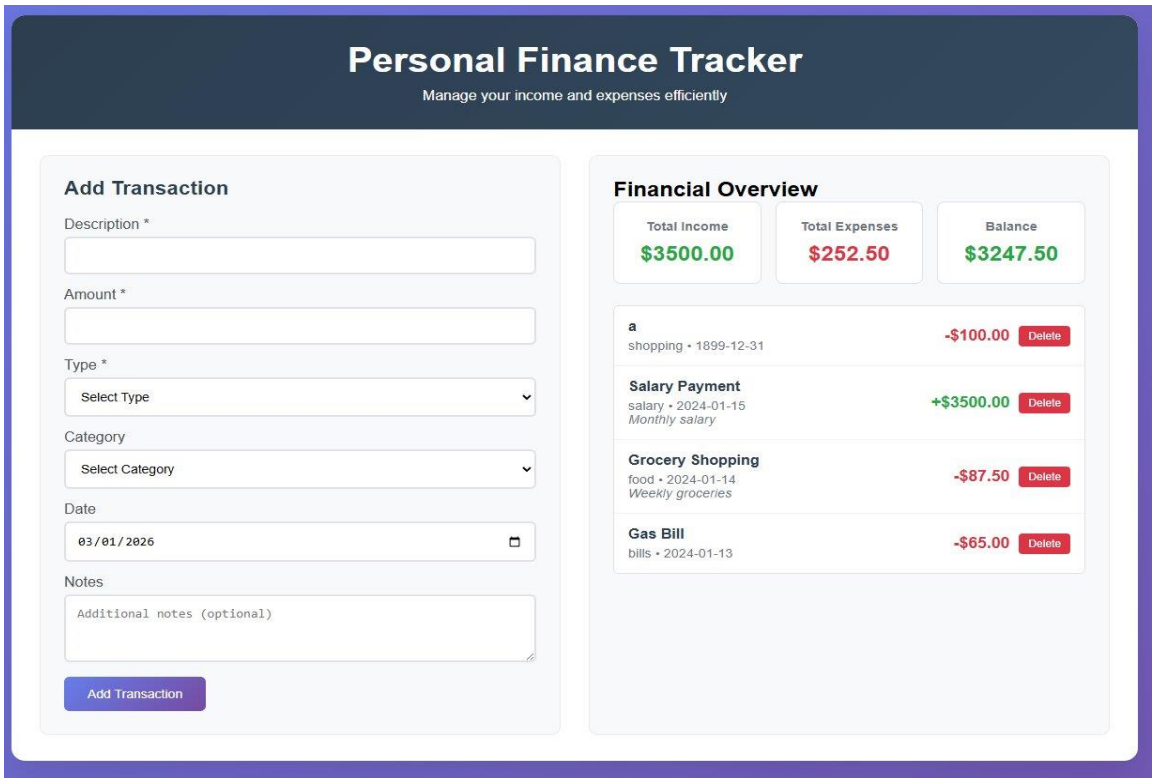

Figure 2 — BUG-002: Date 1899-12-31 accepted; transaction visible in lis

5.3. BUG-003 — Date Above Maximum Boundary (2100-01-01) Is Accepted

Bug ID	BUG-003
Severity	**MEDIUM**
Related TC	TC-022
Component	Date Field Validation
Spec Reference	Section 3.5 — Date Field: Range = 1900-01-01 to 2099-12-31

Steps to Reproduce:

14. Navigate to the application.
15. Enter "a" in the Description field.
16. Enter 100 in the Amount field.
17. Select "Expense" from the Type dropdown.
18. Select "Shopping" from the Category dropdown.
19. Set the Date field to "2100-01-01" via JavaScript injection.
20. Click "Add Transaction".

Expected Result	Actual Result
Transaction is rejected. Date "2100-01-01" is out of the allowed range and should not be stored.	Application accepts the date 2100-01-01 and adds the transaction to the list. The "Adding…" button state is also visible during submission.

Screenshot:

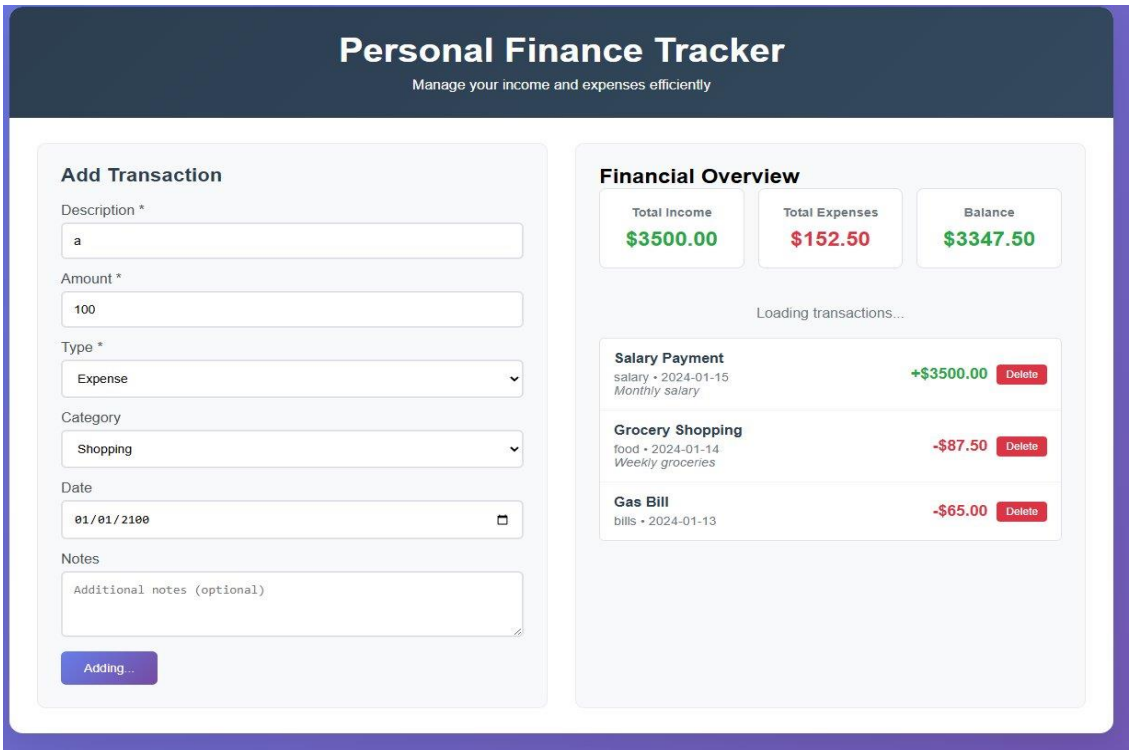

Figure 3 — BUG-003: Date 2100-01-01 accepted; "Adding..." state visible during submission

5.4. BUG-004 — Notes Field Accepts More Than 200 Characters

Bug ID	BUG-004
Severity	**HIGH**
Related TC	TC-025
Component	Notes Field Validation
Spec Reference	Section 3.6 — Notes Field: Maximum Length = 200 characters

Steps to Reproduce:

21. Navigate to the application.
22. Enter "a" in the Description field.
23. Enter 100 in the Amount field.
24. Select "Expense" from the Type dropdown.
25. Select "Shopping" from the Category dropdown.
26. Enter a 201-character string in the Notes field.
27. Click "Add Transaction".

Expected Result	Actual Result
Transaction is rejected. A validation error is shown. No transaction with 201+ character notes is stored.	Application accepts the 201-character note and adds the transaction. The oversized note text also breaks the transaction list layout (see BUG-005).

Screenshot:

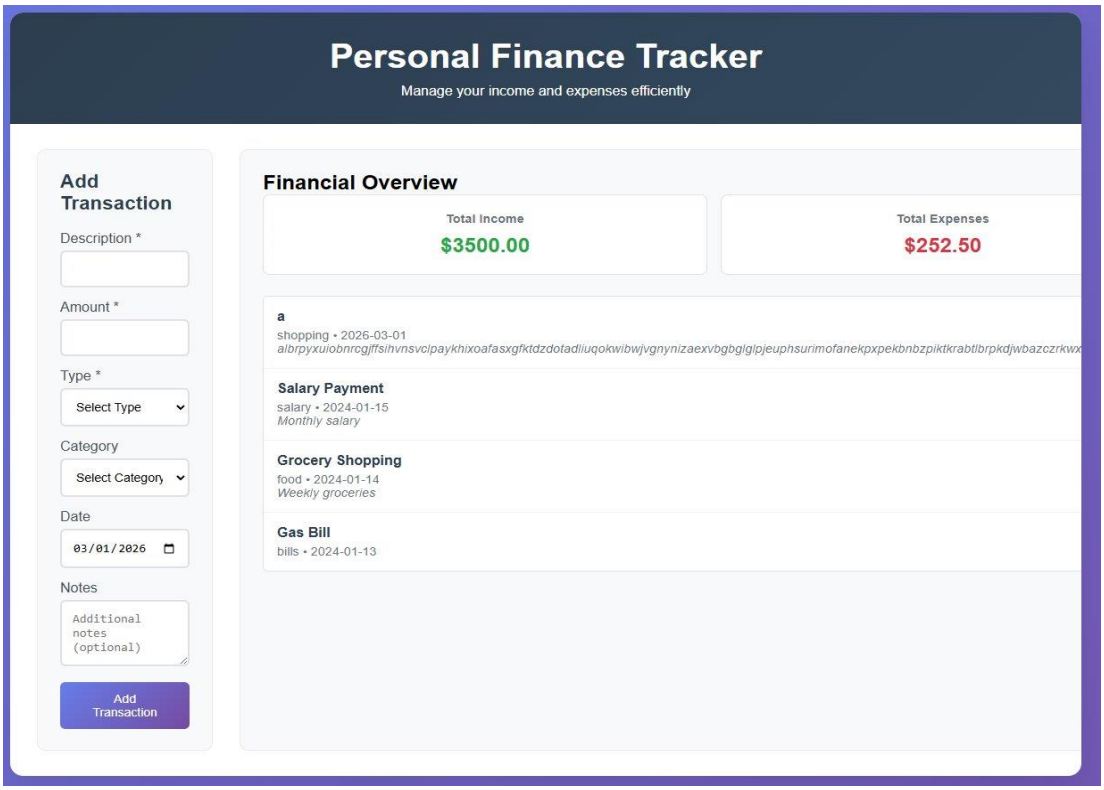

Figure 4 — BUG-004: 201-character note accepted and stored; layout overflow visible

5.5. BUG-005 — Long Text in Description or Notes Causes UI Layout Overflow

Bug ID	BUG-005
Severity	**MEDIUM**
Related TC	TC-004, TC-025
Component	Transaction List UI — Responsive Layout
Spec Reference	Section 5.1 — Responsive Design; Section 5.2 — Visual Design Standards

Steps to Reproduce:

28. Navigate to the application.
29. Enter a 100-character description or a 200+ character note.
30. Add the transaction.
31. Observe the transaction list.

Expected Result	Actual Result
Transaction list renders correctly. Delete button is visible. Balance card is fully visible.	When a description or note is very long, the text overflows the transaction row. The Delete button is pushed off-screen and the Balance amount in the Financial Overview is partially hidden.

Screenshot:

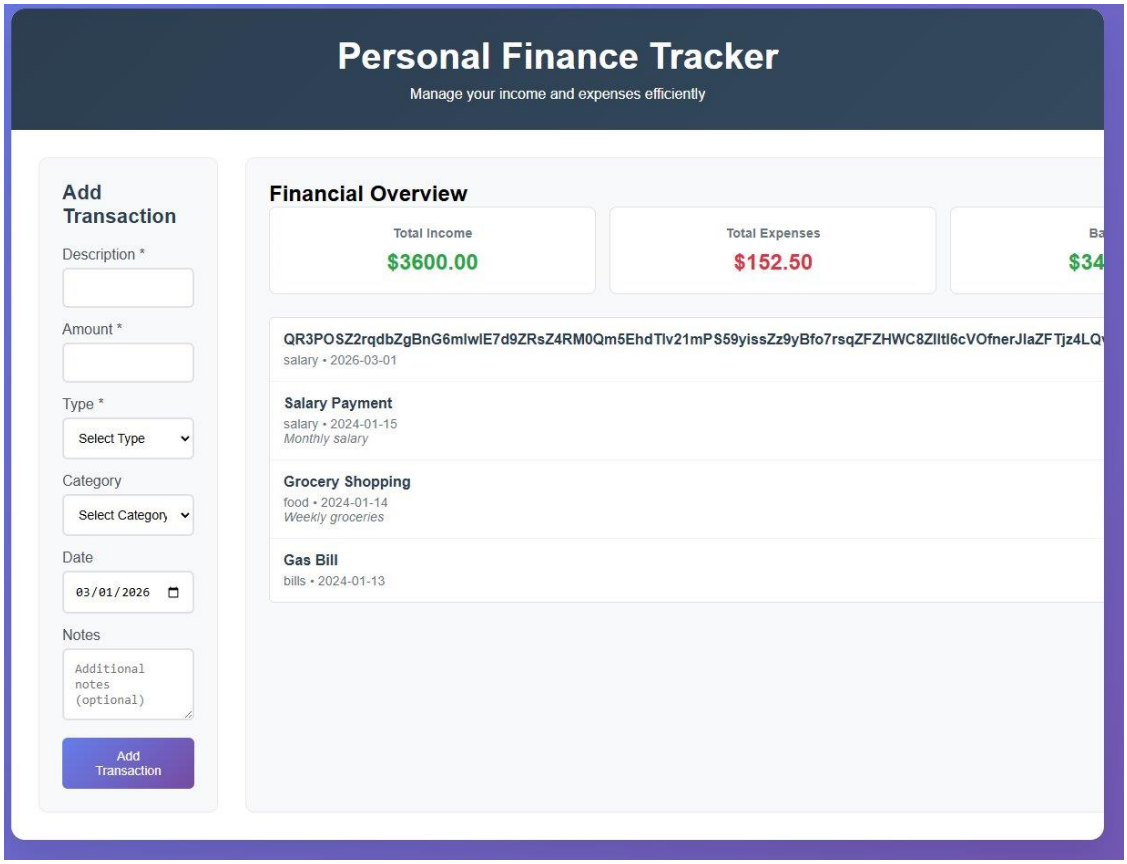

Figure 5 — BUG-005: Long description overflows; Delete button and Balance card are cut off

6. Recommendations

Based on the test results, the following fixes are recommended in priority order:

Bug ID	Priority	Issue	Recommended Fix
BUG-001	**HIGH**	Amount field accepts values above $999,999.99	Add server-side and client-side check: if (amount > 999999.99) show error "Amount seems too large"
BUG-002	**MEDIUM**	Date field accepts dates before 1900-01-01	Add validation: if (date < "1900-01-01") reject and show error
BUG-003	**MEDIUM**	Date field accepts dates after 2099-12-31	Add validation: if (date > "2099-12-31") reject and show error
BUG-004	**HIGH**	Notes field accepts more than 200 characters	Add validation: if (notes.length > 200) show error "Notes must be 200 characters or less"
BUG-005	**MEDIUM**	Long text causes transaction row layout overflow	Apply CSS overflow-wrap: break-word and max-width constraints to transaction description and note elements